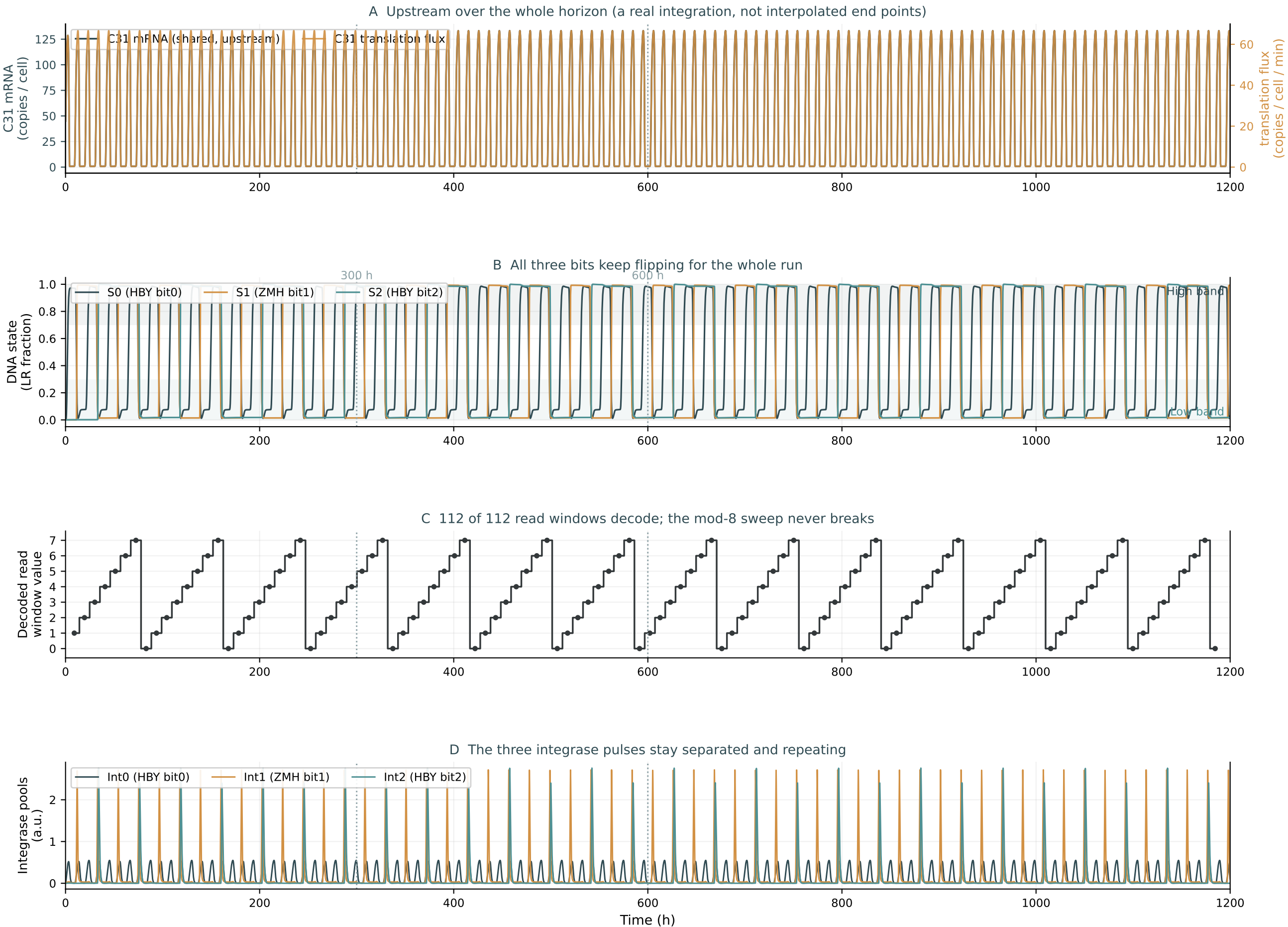


Frozen hybrid HBY bit0 / ZMH bit1 / HBY bit2, one continuous run

1200 h, 1 min grid, DOP853 rtol 2e-07 / atol 2e-09 | clock 0.3/2, n_A1_gate 6, uM_per_au 5.75
certified_v1 = True at every probed horizon | steady read windows 104 (mod-8 increments True, boundary clips 0) | min timing margin 1.1785 h



Model: HBY bit0 + ZMH current-code A0/F0 and bit1 + HBY A1/F1 and bit2, 34 states
Input: Han v53d upstream, true integration to the full horizon (no end-point interpolation, normalisation window unchanged)
Horizon: 300 / 600 / 1200 h from a single run; the 300 h slice reproduces the frozen certification exactly